

KUCHING INTL, Malaysia

WMO: 964130

Lat: 1.485N

Lon: 110.347E

Elev: 89

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	71.8	72.7	69.3	107.9	82.3	70.0	110.6	81.1	14.2	81.4	12.8	81.6	2.5	140	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	14.9	93.4	79.2	92.4	79.1	91.5	79.1	81.8	89.3	81.0	88.5	80.4	87.8	5.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.6	154.0	86.1	79.0	151.2	85.3	78.7	149.3	84.7	45.7	89.9	44.8	89.2	44.2	88.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.7	9.4	70.4	95.9	1.1	1.2	69.6	96.8	68.9	97.5	68.3	98.2	67.4	99.1		
(5)			69.1	83.7	1.1	1.8	68.3	85.0	67.6	86.1	67.0	87.1	66.1	88.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	79.8	79.9	81.1	82.1	82.7	82.4	82.2	82.2	81.8	81.4	81.3	80.6
	DBStd	2.00	1.95	1.78	1.73	1.54	1.72	1.85	1.97	2.05	1.96	1.70	1.42	1.69
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11490	925	838	965	964	1015	971	997	998	954	975	938	950
	CDD65	6015	460	418	500	514	550	521	532	533	504	510	488	485
	CDH74	51821	3488	3344	4281	4525	5232	4869	4999	4990	4404	4101	3845	3743
	CDH80	18367	892	888	1390	1650	2035	1914	2027	2037	1676	1448	1292	1118
Wind	WSAvg	3.6	3.8	3.8	3.5	3.2	3.3	3.4	3.7	3.8	3.6	3.5	3.4	3.7
Precipitation	PrecAvg	165.7	27.0	19.9	13.8	11.3	10.0	9.0	7.8	9.4	10.2	13.5	14.0	19.2
	PrecMax	208.3	53.2	61.4	29.4	22.2	17.0	21.3	17.6	21.5	17.8	24.5	24.0	36.9
	PrecMin	109.1	5.7	4.7	5.9	3.3	3.4	3.9	1.9	0.9	3.7	5.6	6.2	5.7
	PrecStd	21.3	11.5	9.9	5.5	4.1	2.8	3.3	3.3	4.3	3.6	4.6	4.1	5.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	89.8	91.6	93.1	93.6	93.6	94.7	94.8	93.5	93.1	91.7	91.4
		MCWB	79.2	78.7	78.9	79.9	80.1	79.1	79.2	78.8	79.2	78.9	79.2	79.3
	2%	DB	88.0	88.0	89.9	91.4	92.5	92.0	92.7	92.9	91.7	91.2	89.9	89.4
		MCWB	78.7	78.4	78.8	79.6	79.9	79.1	78.8	78.5	79.1	79.0	78.9	78.7
	5%	DB	86.3	86.4	88.2	89.8	91.2	91.1	91.2	91.3	90.0	89.4	88.5	87.8
		MCWB	77.9	77.8	78.7	79.2	79.7	79.1	78.7	78.3	78.8	78.7	78.8	78.6
	10%	DB	84.6	85.3	86.9	88.1	89.5	89.4	89.4	89.6	88.5	87.6	87.3	86.1
		MCWB	77.3	77.4	78.4	79.1	79.6	78.9	78.3	78.2	78.6	78.6	78.6	78.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.4	80.2	81.3	81.8	83.2	82.4	81.9	81.0	82.3	81.0	81.1	80.9
		MCDB	87.0	86.6	87.8	89.7	90.7	90.2	90.1	89.2	89.5	88.9	88.2	87.5
	2%	WB	79.3	79.2	80.0	80.8	81.6	80.9	80.6	80.1	80.6	80.1	80.0	79.8
		MCDB	85.1	85.0	86.7	88.2	89.1	88.4	88.6	88.3	88.4	87.4	87.0	86.4
	5%	WB	78.8	78.7	79.2	80.0	80.7	80.0	79.7	79.5	79.7	79.4	79.2	79.1
		MCDB	84.1	84.1	85.5	86.9	87.9	87.3	87.5	87.5	87.2	86.5	85.9	85.2
	10%	WB	78.1	78.1	78.7	79.3	79.9	79.3	79.0	78.9	79.0	78.8	78.7	78.5
		MCDB	83.0	83.1	84.6	85.8	86.7	86.4	86.5	86.7	86.2	85.7	85.2	84.2
Mean Daily Temperature Range	5% DB	MDBR	11.2	11.4	13.0	14.5	14.9	15.1	15.5	15.7	15.0	14.5	14.1	12.7
		MCDBR	13.4	13.4	15.0	16.1	16.8	17.0	17.5	17.7	17.1	16.6	15.5	14.6
		MCWBR	5.2	5.2	5.8	6.1	6.3	6.3	6.3	6.4	6.7	6.4	6.1	5.8
	5% WB	MCDBR	12.3	12.1	13.8	15.0	15.5	15.6	16.1	16.4	15.8	15.2	14.5	13.5
		MCWBR	5.2	5.2	5.8	6.2	6.4	6.4	6.6	6.5	6.9	6.4	6.1	5.8
Clear-Sky Solar Irradiance	taub	0.422	0.430	0.430	0.415	0.420	0.418	0.438	0.527	0.565	0.549	0.427	0.413	
	taud	2.450	2.406	2.398	2.441	2.428	2.443	2.381	2.111	2.004	2.040	2.419	2.478	
	Ebn at Noon	287	286	284	283	274	271	267	249	245	250	284	288	
	Edh at Noon	37	39	40	37	36	35	38	51	58	56	38	36	
All-Sky Solar Radiation	RadAvg	1303	1394	1568	1610	1575	1532	1567	1560	1441	1410	1436	1365	
	RadStd	126	96	82	62	69	63	93	82	139	176	75	88	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.67	+0.41	+0.40	+0.99	+1.06	+1.15	N/A	N/A	+275	+275
(47) Regional (1 neighbor)		+0.67	+0.41	+0.40	+0.99	+1.06	+1.15	N/A	N/A	+275	+275

Nomenclature: See separate page